

Hospital Readmission Reduction Analytics

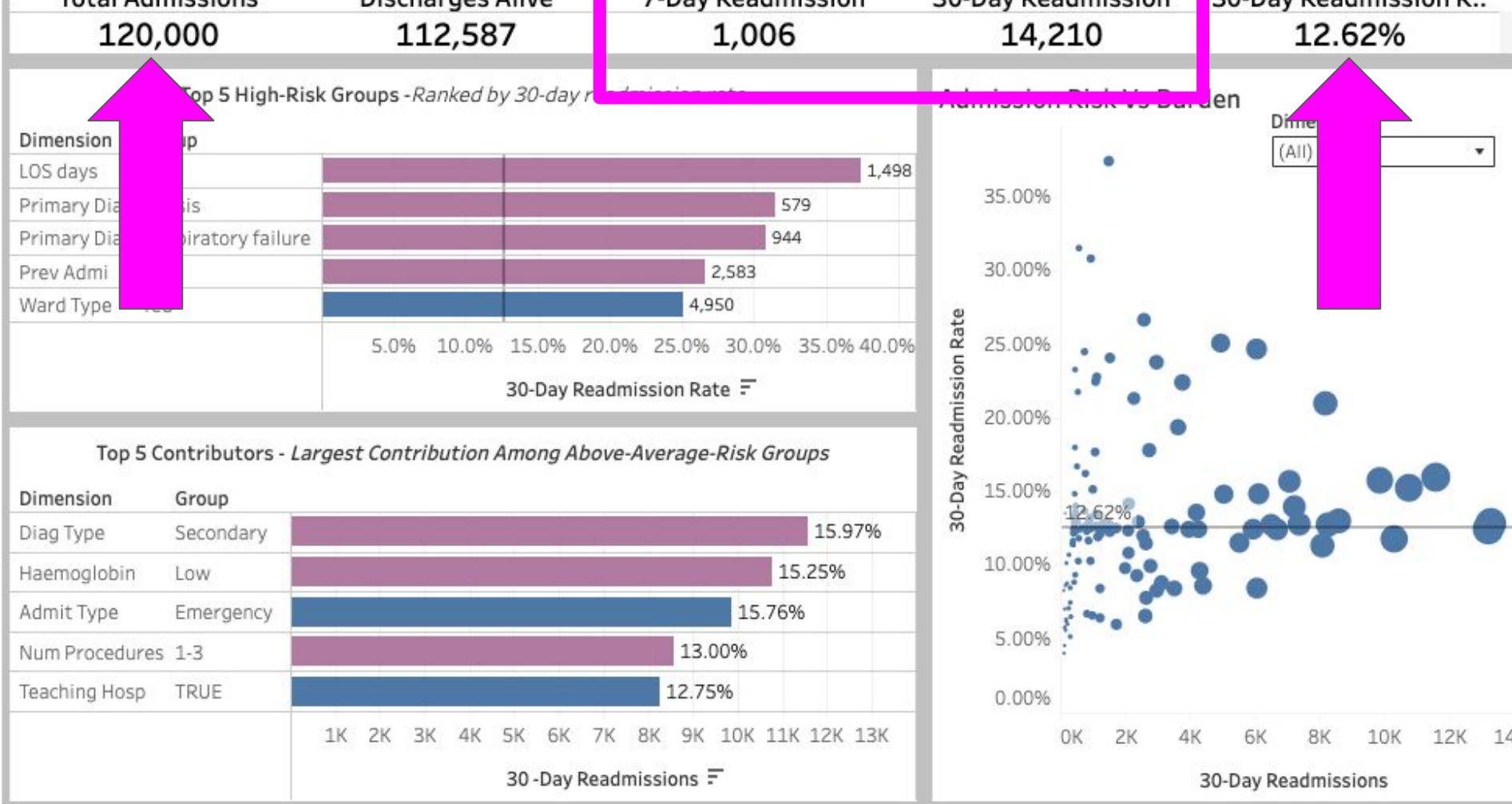
From population-level burden to patient-level prioritization

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Executive Overview

How Are We Doing?

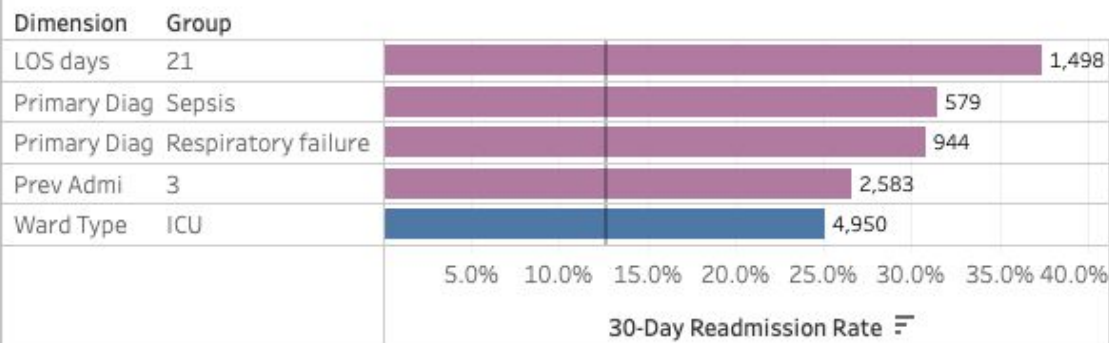
Analysis Period: Jan 2015 - Dec 2024



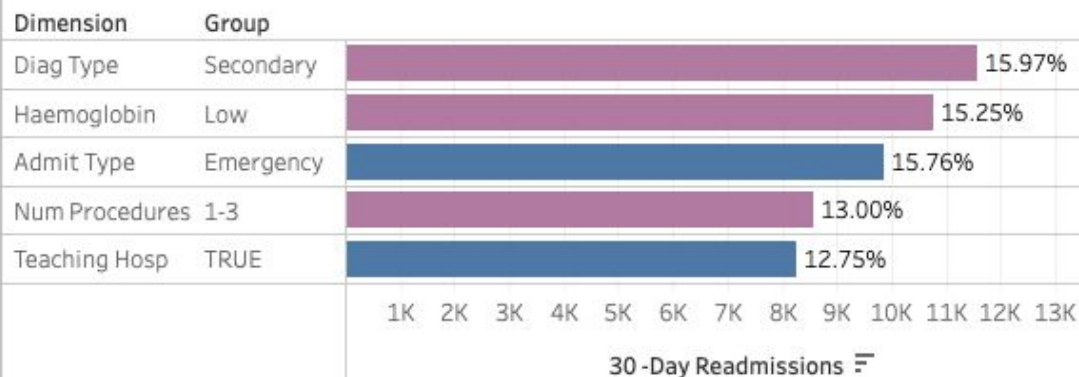
Where Should the Hospital Focus?

Risk Vs Burden

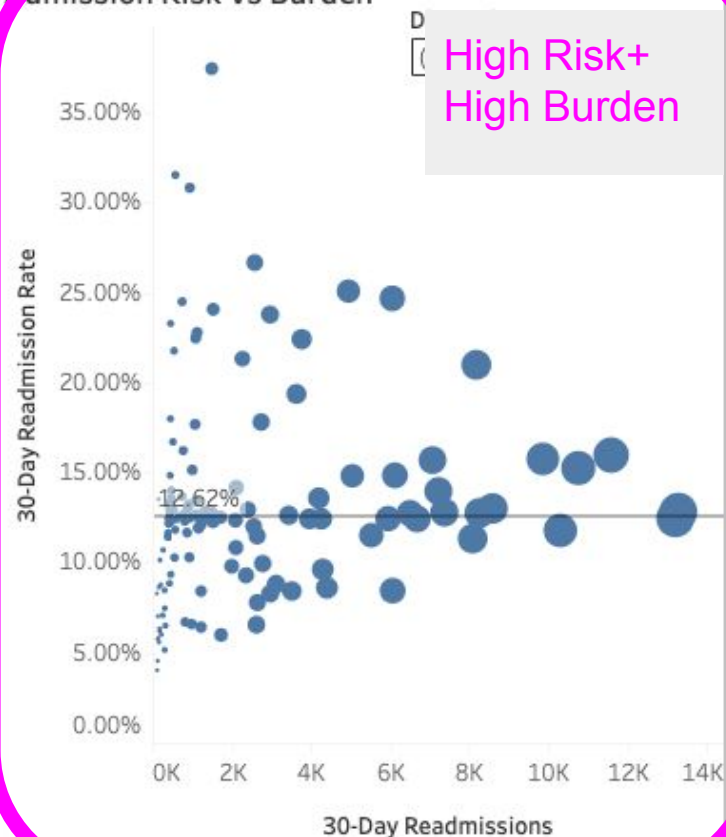
Top 5 High-Risk Groups - Ranked by 30-day readmission rate



Top 5 Contributors - Largest Contribution Among Above-Average-Risk Groups

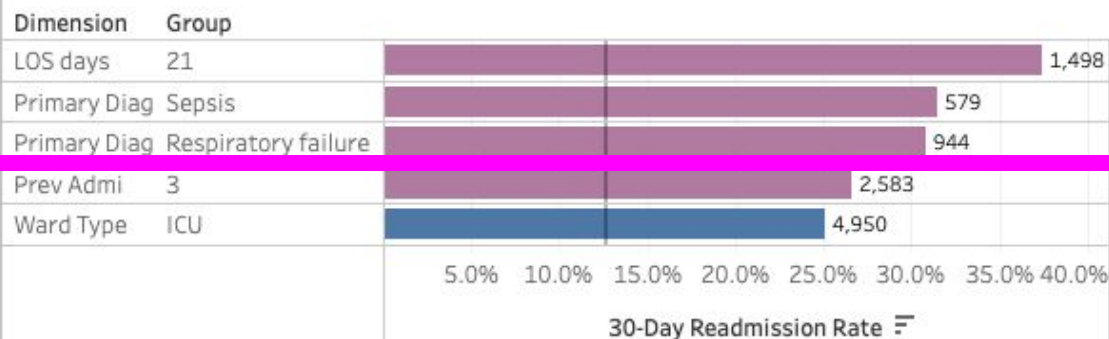


Admission Risk Vs Burden

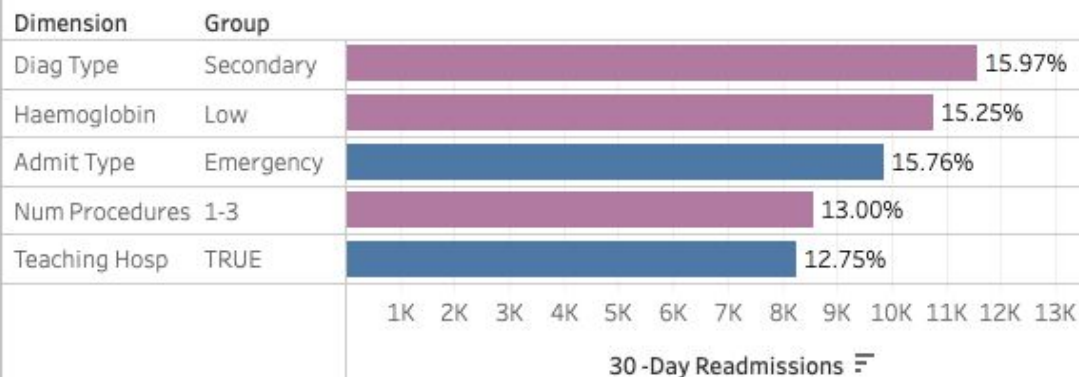


Risk Vs Burden

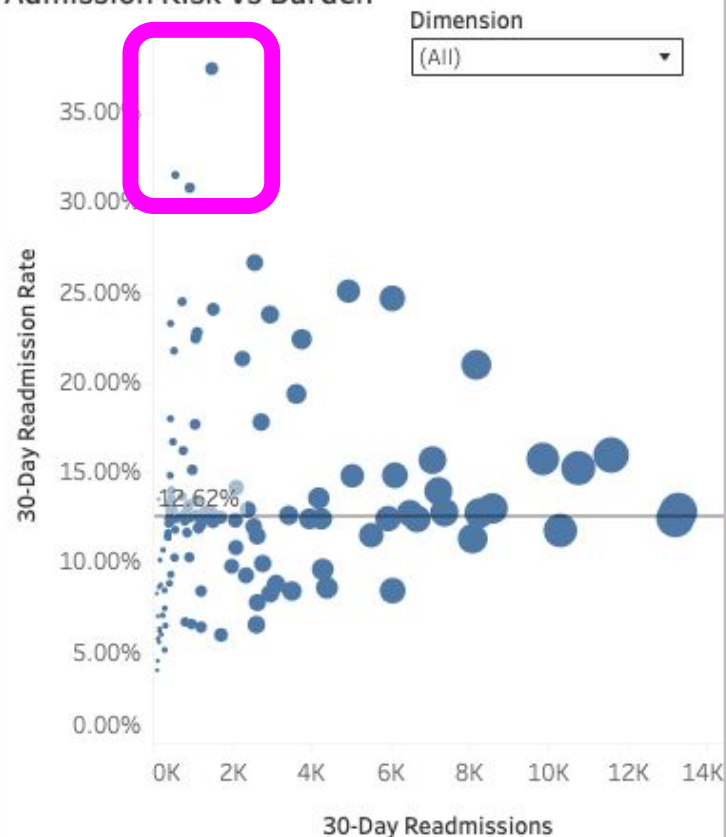
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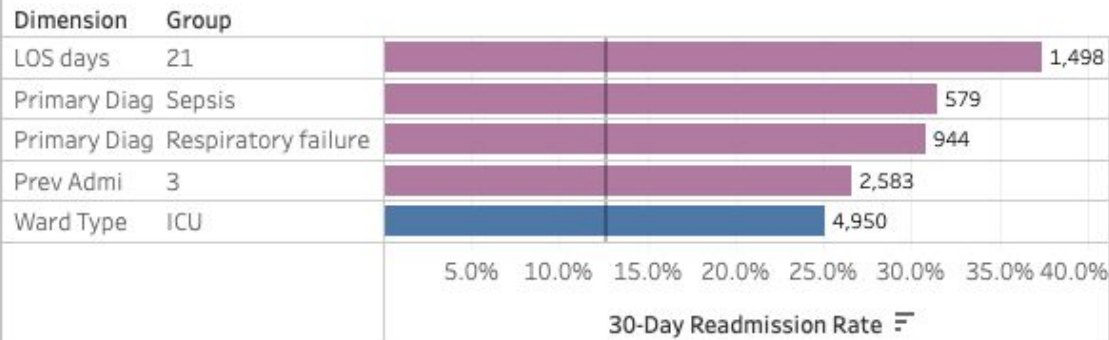


Admission Risk Vs Burden

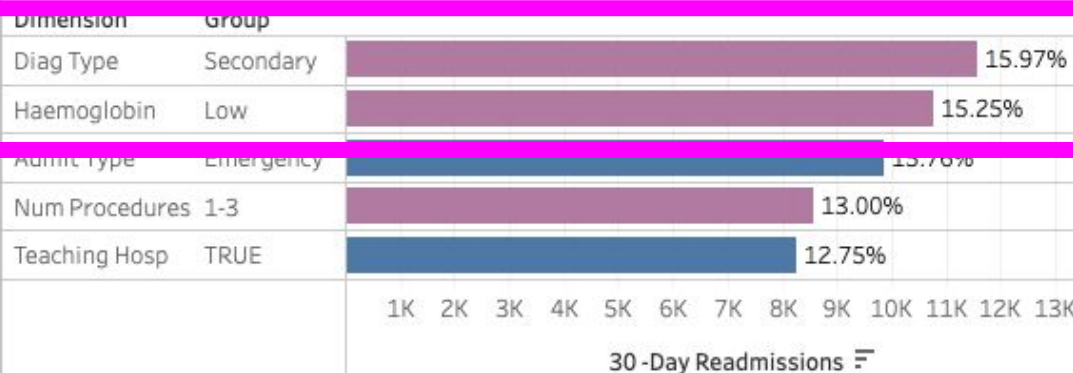


Risk Vs Burden

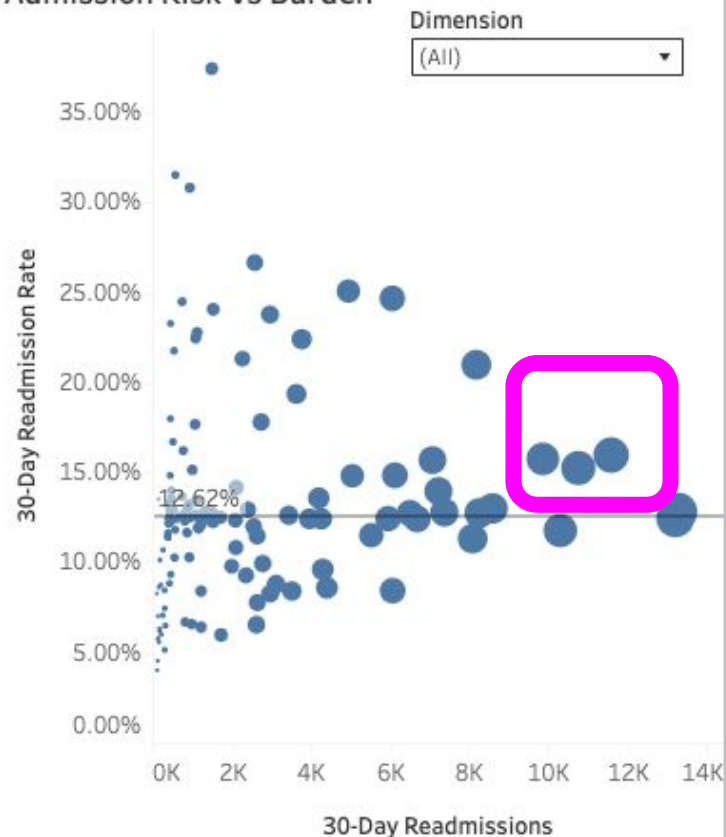
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Top 5 Contributors - Largest Contribution Among Above-Average-Risk Groups



Admission Risk Vs Burden



What Is Associated With Readmission Burden?

Profile of Top Readmission Contributor

Secondary Diagnosis

Total Admissions
77,128

% of Total Admisson
64.27%

Discharges Alive
72,532

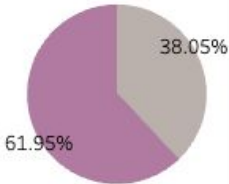
30-Day Readmission
11,584

30-Day Readmission Rate
15.97%

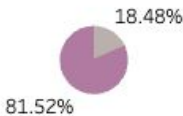
Primary Vs Secondary Diagnosis

Read.. 2

Not
Readmitted

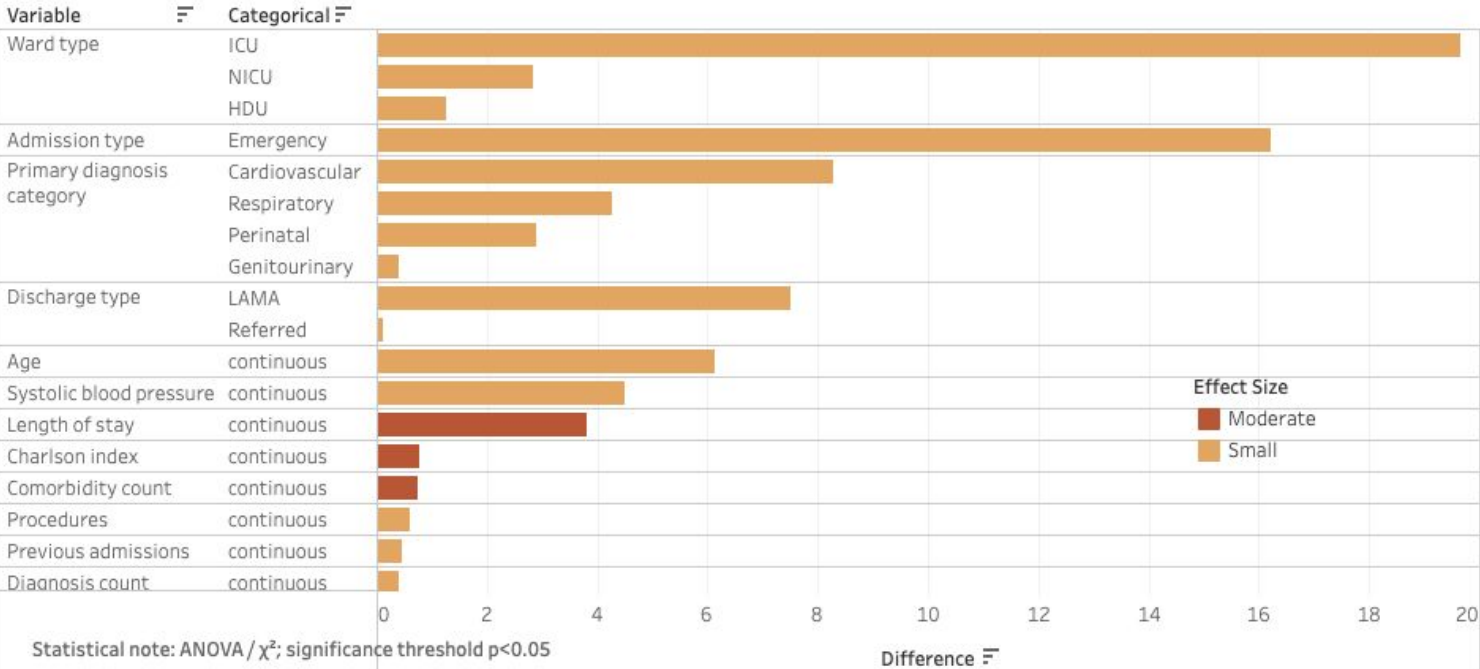


Readmitted



Significant Factors Associated with Readmission

Statistically significant factors are shown with effect size to indicate magnitude.



Predictive Modeling : Enables Patient-Level Prioritization

Can We Prioritize Patients When Resources Are Limited?

Model comparison and Selection

Model	Test AUC
Logistic Regression — baseline* (Selected)]	0.7545
Logistic + 34 ICD-10 indicators	0.7554
Logistic + ICD-10 categories	0.7548
Logistic + diagnosis burden	0.7544
XGBoost	0.7498
Random Forest	0.7331–0.7408

Percentile-based risk stratification

Highest-risk percentile	Patients targeted	Observed readmissions captured	Share of all test-set readmissions	Observed readmission rate
Top 5%	1,126	545	19.2%	48.4%
Top 10%	2,252	916	32.2%	40.7%
Top 15%	3,378	1,199	42.2%	35.5%
Top 20%	4,504	1,441	50.7%	32.0%
Top 25%	5,630	1,637	57.6%	29.1%

Recommendation:

Potential interventions for high-risk patients

- enhanced **discharge planning** and medication reconciliation;
- **follow-up appointments** scheduled before discharge;
- **Post-discharge follow-up** – Call or email
- early outpatient or primary-care follow-up;
- pharmacist or care-manager review;
- **patient education** and teach-back to confirm understanding of discharge instructions;
- coordination of **transportation, medication access**, and other barriers to follow-up;
- targeted **transitional-care or case-management programs** for patients with particularly high risk.

Three questions answered

Where is the problem? What is associated with it?

High-burden and high-rate populations were identified. – Multiple clinical and operational characteristics showed associations.

Who should we prioritize?

The predictive and risk stratification model concentrated observed readmissions within higher-risk patient groups.

What remains to be answered?

Which interventions actually reduce readmissions.-- Carefully Design pilot targeted-care intervention and evaluation.

Thank you